LINEAR REGRESSION PROJECT 1

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1. Introduction

In this report, concentrations of plasma beta-carotene (ng/ml) in human individuals will be studied in relation to personal characteristics and dietary factors. The relation will be modeled using linear regression with beta-carotene levels as the dependent variable and the following 11 explanatory variables, given in Table 1 below.

Variable	**Description**
Age	(In years)
Sex	(Male or female)
Smoking	Never/former/current smoker
BMI	Body mass index, weight/height^2
Vitamins	Vitamin use (fairly often/not often/no)
Calories	Daily calories consumed (MJ)
Fat	Daily fat consumed (g)
Fiber	Daily fiber consumed (g)
Alcohol	Number of alcoholic drink consumed weekly
Cholesterol	Daily consumed cholesterol (mg)
Beta-carotene	Daily dietary beta-carotene consumed (mg)

Table 1. Explanatory variables with short descriptions.

To model this relationship, data of the plasma beta-carotene levels and of the variables in Table 1 have been documented in 315 individuals. The linear model is given by

$$Y_i = \mathbf{x}_i\beta + \epsilon_i, \quad \epsilon_i \in N(0, \sigma^2), \quad i = 1, 2, 3, ...12,$$

where Y_i is the beta-carotene levels, $\mathbf{x}_i = (x_{i1}, x_{i2}, ..., x_{ip})$ is the i'th observation of the p'th dependent variable, and the error terms ϵ_i are pairwise independent.

2. Plasma beta-carotene levels in relation to BMI

In this first section, plasma beta-carotene levels will be studied as a function of only one dependent variable, the BMI.

2.1. **a).** We begin by comparing two different models, *Lin* and *Log*, where

$$Lin : \text{betaplasma}_i = \beta_0 + \beta_1 \text{bmi}_i + \epsilon_i, \quad \epsilon_i \in N(0, \sigma^2), \quad i = 1, ..., n, \tag{2.1}$$

and

$$Log : \ln(\text{betaplasma})_i = \beta_0 + \beta_1 \text{bmi}_i + \epsilon_i, \quad \epsilon_i \in N(0, \sigma^2), \quad i = 1, ..., n. \tag{2.2}$$

The residuals of the *Lin* and *Log* models are shown in Figure 1 and Figure 2 respectively, below.

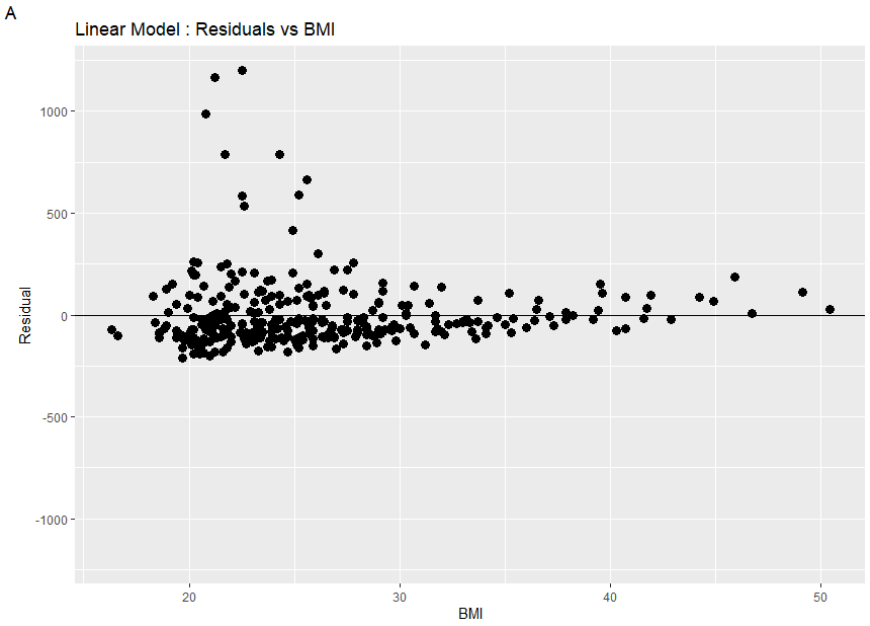


FIGURE 1. Plot of residuals vs BMI in the linear model.



FIGURE 2. Plot of residuals vs BMI in the logarithmic model.

Comparing the residual plots in Figures 1 and 2, we observe that the residuals of the linear model exhibit considerable asymmetry, with several unusually large positive residuals, particularly at lower BMI values. In contrast, the residuals of the logarithmic model appear more symmetrically distributed around zero, with a less pronounced pattern of extreme positive deviations.

Although some variation in the residual spread remains, the logarithmic transformation appears to improve the residual distribution and provides better support for the assumptions of the linear regression model. We therefore consider the logarithmic model more appropriate for the subsequent analysis.

2.2. **b).** We now show estimates of β_0 and β_1, pertaining to the intercept and BMI variable in the *Log* model given in (2.2), including 95% confidence intervals. The estimates are given in Table 2 below.

Estimate	95% **Confidence interval**
$\hat{\beta_0} = 5.89$	$I_{\beta_0} = (5.53, 6.25)$
$\hat{\beta_1} = -0.04$	$I_{\beta_1} = (-0.05, -0.02)$

TABLE 2. Table of beta parameters estimates and 95% normal confidence intervals.

Next, in Figure 3 we display a plot of the log beta-carotene against BMI along with the estimated *Log* model line (as in (2.2)), a 95% confidence interval for the estimate, and also a 95% prediction interval for future observations.

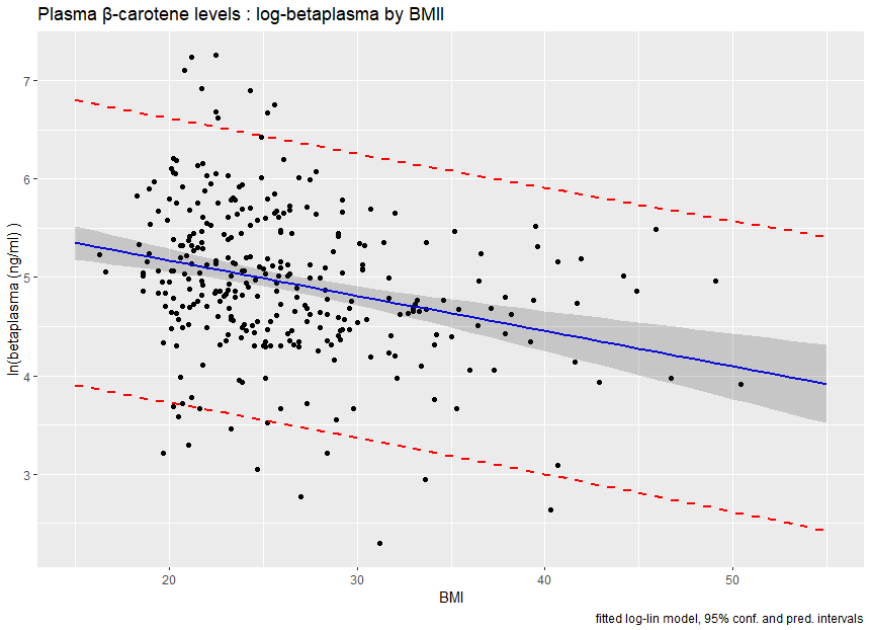


FIGURE 3. Plot of log beta-plasma vs BMI, estimated *Log* model line, 95% estimate for the line, and a 95% prediction interval.

Finally, we transform the data that was plotted in log-scale in Figure 3 back to linear scale and plot the result, as shown in Figure 4 below.

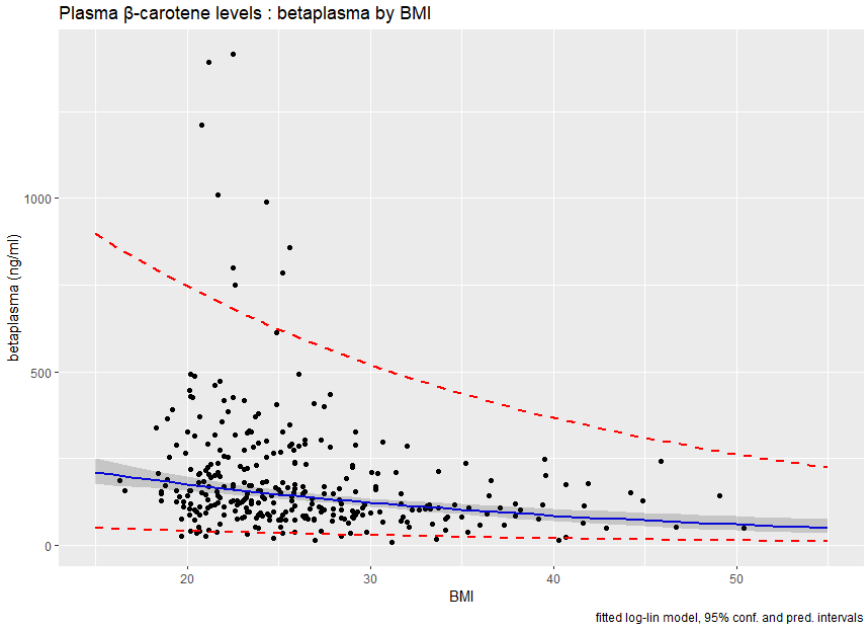


FIGURE 4. Plot of beta-plasma vs BMI, estimated *Lin* model line, 95% estimate for the line, and a 95% prediction interval.

2.3. **c).** We now look at how the expected plasma beta-carotene levels (given in ng/ml) change as the BMI increases or decreases, according to the *Log* model.

The beta-carotene levels decay exponentially with respect to BMI, with a decay factor of 0.965 per unit change in BMI (with one unit being $1kg/m^2$), given by e^{β_1}. A 95% confidence interval for this rate of change is given by (0.952, 0.978), while holding other factors constant, corresponding to an approximate 3.5% decrease per unit change.

Holding other variables constant, a one unit decrease ($e^{-\beta_1}$) in BMI is associated with a 3.65% increase in beta-carotene levels on average. The multiplicative factor is 1.037 and a 95% confidence interval for this change rate is given by (1.023, 1.051).

Holding other variables constant, a ten unit decrease ($e^{-10\beta_1}$) in BMI is associated with a 43.17% increase in beta-carotene on average, with a multiplicative factor of 1.43. A 95% confidence interval for the change rate is given by (1.251, 1.639).

A compact summarization of the above numbers is given in Table 3 below.

Unit change in BMI	**Expected change rate (ng/ml)**	**Confidence interval**
+1 unit	0.965	(0.952, 0.978)
-1 unit	1.037	(1.023, 1.051)
-10 units	1.43	(1.251, 1.639)

TABLE 3. Table of expected rate of change in beta-carotene levels (ng/ml) with pertaining 95% normal confidence intervals, with respect to different unit changes in BMI.

2.4. **d).** Next, we want to test whether there is a significant linear relationship between the log plasma beta-carotene levels and BMI, again with respect to the *Log* model. We do this on a significance level of

$\alpha = 5\%$, using a t-test. Following are the hypotheses

$$H_0 : \beta_1 = 0,$$
$$H_1 : \beta_1 \neq 0.$$

With the help of R, we see that we can reject the null hypothesis at a 5% significance level (as the p-value is (much) lower than 0.05) and claim that BMI *does* have an effect on beta-carotene levels. In Table 2 below, the value of the test-statistic (t-value) when H_0 is true, its degrees of freedom (d.f.) and the p-value are given. Note that, under the null hypothesis, the test-statistic follows a t-distribution with 313 degrees of freedom.

t-value	**d.f.**	**p-value**
-5.2298	313	$3.1108 \cdot 10^{-7}$

TABLE 4. Table containing the test-value, degrees of freedom and the p-value for testing whether there is a significant ($\alpha = 5\%$) linear relationship between plasma beta-carotene levels and BMI.

3. Plasma beta-carotene and smoking habits

The effects of smoking habits on beta-carotene levels are now going to be studied.

3.1. **a).** We begin by presenting the three different categories of smoking habits in the 315 studied individuals and some their pertaining data. This is shown in Table 5 below.

Category	**N**	**Mean**	**std. dev.**	**log mean**	**log std. dev.**
Never	157	206	193	5.05	0.745
Former	115	193	192	4.94	0.798
Current	43	121	78.8	4.61	0.624

TABLE 5. Table containing the smoking status of 315 (total N) individuals with pertaining mean, standard deviation (std. dev.), log mean and log standard deviation of the beta-plasma (ng/ml) for each category.

Seeing as the "Never" category contains the most observations, it is reasonable to choose this category as reference category. This decision is further motivated by the natural division between the "Never" category and the "Former" and "Current" categories, where in the latter the individuals have at some point smoked, unlike those in the first category.

We now do a visual comparison between a standard linear model and a logarithmic model for the beta-carotene levels against each smoking status category. In Figure 5 and Figure 6 are violin-plots for the beta-carotene and log beta-carotene against the smoking status respectively.

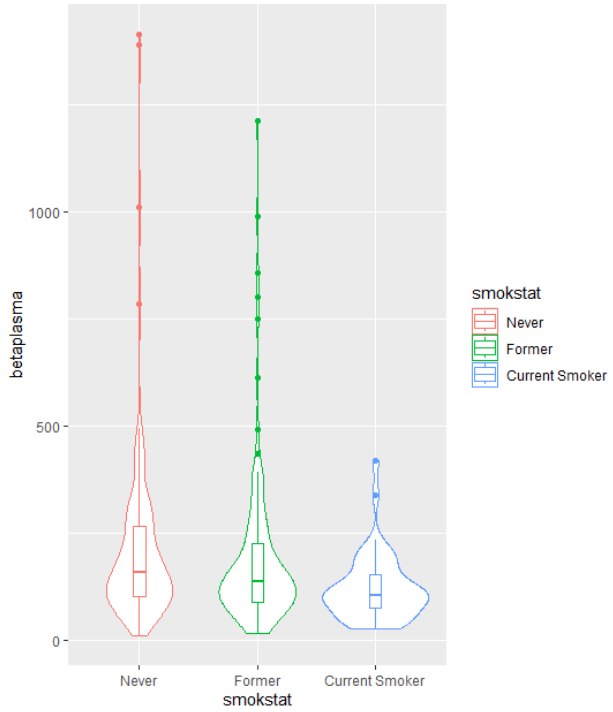


FIGURE 5. Violin-plot of beta-plasma vs smoking status.

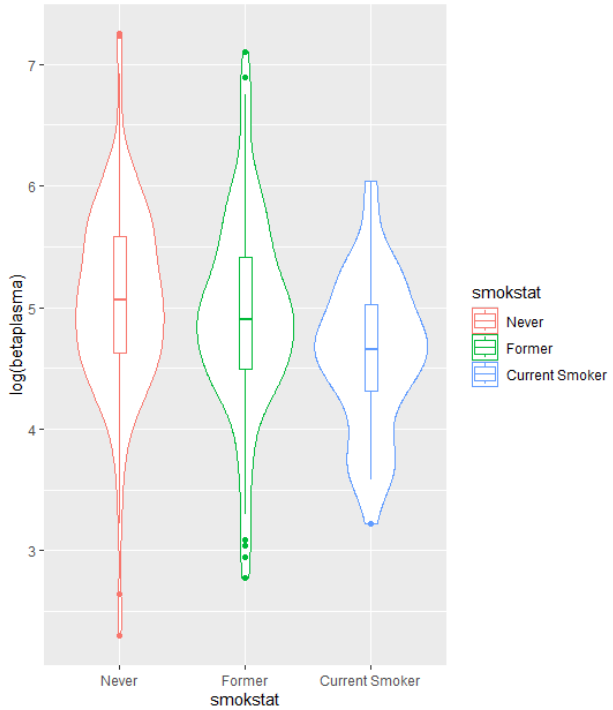


FIGURE 6. Violin-plot of log beta-plasma vs smoking status.

Similarly to Section 2, there is a noticeable difference between the models; the *Log* model has a lower variance. This is seen by how the data is spread out in the two different plots for the different smoking categories.

3.2. **b).** We now let the beta-carotene levels depend on the categorical variable related to the smoking status and create two *Log* models; one with the "Never" category as reference (Model 1) and one with the "Current" as reference (Model 2).

In the Log-Lin model with "Never" as the reference category (Model 1), the dummy variable x_{1i} equals to 1 for "Former" smokers and 0 otherwise, and is connected to the β_1 parameter. The dummy variable x_{2i} for "Current" smokers works in a similar way and is connected to β_2. The "Never" category is captured by the intercept term β_0 when both dummy variables equal 0. In Model 2, the intercept β_0 is connected to the "Current" smokers category, the β_1 is connected to the non-smokers category, and the β_2 is connected to the "Former" smokers category. In Table 6 and Table 7, estimates of the different β's are given for Model 1 and Model 2 respectively.

	Estimate	**std. deviation**
$\hat{\beta_0}$	5.051	0.060
$\hat{\beta_1}$	-0.110	0.092
$\hat{\beta_2}$	-0.437	0.130

TABLE 6. Estimates of the β-values in Model 1.

	Estimate	**std. deviation**
$\hat{\beta_0}$	4.614	0.114
$\hat{\beta_1}$	0.437	0.129
$\hat{\beta_2}$	0.327	0.134

TABLE 7. Estimates of the β-values in Model 2.

The category "Never" is considerably larger than the "Current" category (157 vs 43 individuals), which is reflected in the tables of the β-estimates; Model 1 with the larger category has much smaller standard deviations. Thus, Model 1 is a more suitable choice of model.

3.3. **c).** We now want the expected beta-carotene levels and expected log levels of beta-carotene levels given each smoking category, with pertaining 95% confidence intervals. The expected values for both Model 1 and Model 2 are the same, because both models estimate the same three group means in the same linear combination, just with different parameterizations and dummy variables that are used to shift the reference point. Thus, only two tables will be needed to present the values. In Table 8 and Table 9 are the log beta-carotene values and the beta-carotene values, respectively.

Category	**Expected log beta-carotene**	95% **confidence interval**
Never	5.051	(4.933, 5.169)
Former	4.941	(4.803, 5.079)
Current	4.614	(4.389, 4.839)

TABLE 8. Expected log beta-carotene (ng/ml) and 95% confidence intervals for each category, for Model 1 and 2.

Category	**Expected beta-carotene**	95% **confidence interval**
Never	156.155	(138.804, 175.676)
Former	139.928	(121.936, 160.574)
Current	100.850	(80.525, 126.306)

TABLE 9. Expected beta-carotene (ng/ml) and 95% confidence intervals for each category, for Model 1 and 2.

The sizes of the expected values in Table 8 and Table 9 can be related to the sizes of the means in Table 5 by the Law of Large Numbers.

3.4. **d).** Continuing with Model 1, we now want to test, with a significance level of $\alpha = 0.05$, whether there are significant differences in log beta-carotene levels between any of the smoking categories ("Never", "Former" and "Current"). To do this, we use an F-test with hypotheses by

$$\mathrm{H}_0 : \beta_1 = \beta_2 = 0,$$
$$\mathrm{H}_1 : \beta_1 \neq 0 \text{ and/or } \beta_2 \neq 0.$$

The observed value of the F-test is 5.75. If the Null Hypothesis is true, then the test statistic follows an F-distribution with 2 and 312 degrees of freedom ($\mathrm{F} \in \mathrm{F}(2, 312)$). The p-value is given by 0.00353. Therefore, we can reject the Null Hypothesis at a level of significance $\alpha = 5\%$ and conclude that smoking status is relevant to beta-carotene levels.

4. Multiple linear regression

In the two previous sections we have studied relations between beta-carotene levels and BMI and smoking using simple linear regression. We now want to continue study how beta-carotene levels varies according to the explanatory variables given in Table 1 using *multiple linear regression*.

4.1. **a).** We begin by presenting two frequency tables; Table 10 contains the division of the 315 participants in the study between male and females, and Table 11 shows the vitamin use frequency of the participants.

	Female	**Male**
N	273	42

TABLE 10. Frequency table for number (N) of females and males that participated in the study.

	No	**Yes, fairly often**	**Yes, often**
N	111	122	82

TABLE 11. Frequency table for vitamin use by the participants in the study, where N gives the number of observations for each given column.

Seeing as the Female category is almost eight times the size as the Male category, it is a suitable choice as reference variable. This will minimize the variance of any estimate we want to do with data involving this explanatory variable. The reference category for vitamin use may reasonably be chosen as the "No" category. This is since there is no big difference between the frequency between the "No" and the "Yes, fairly often" categories, while the "No" category has a more natural status as reference variable.

4.2. **b).** Next, we calculate the pairwise correlations between the continuous explanatory variables, i.e., between BMI, Age, Calories, Fat, Cholesterol, Fiber, Alcohol and (dietary) Beta-carotene. The pairs with correlations ± 6 or higher are presented in Figure 7.

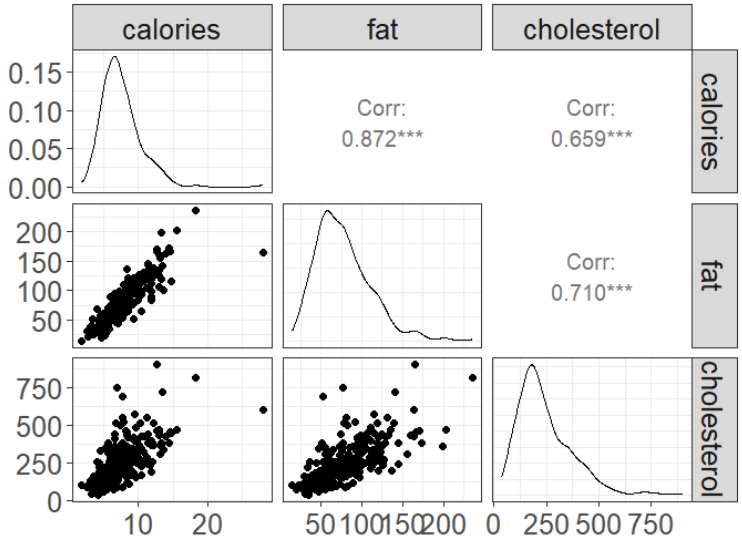


FIGURE 7. Plot of Calories, Fat and Cholesterol against each other.

In the Alcohol variable, there is one outlier; one of the participants drinks remarkably more than the others. This might skew the data towards not finding high correlations between the Alcohol variable and other variables (as this one data point may "lower" the linear relation between alcohol consumption and other variables). In relation to this participant's alcohol consumption, we might also expect a higher consumption of calories. This turns out to be a correct prediction and may be seen in Figure 8.

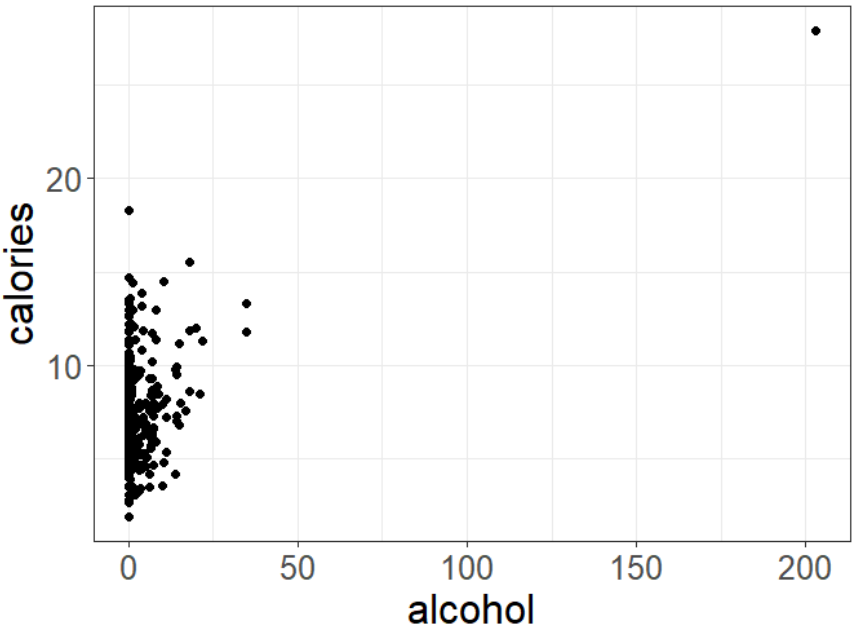


FIGURE 8. Plot of Calories against Alcohol.

4.3. **c).** We now fit a logarithmic model, where logarithmic beta-carotene depends on all of the 11 explanatory variables given in Table 1. The Variance Inflation Factor (VIF) measures how much the variance of an estimated regression coefficient increases due to collinearity, calculated as:

$$\text{VIF} = \frac{1}{1 - R^2}$$

where R^2 is the coefficient of determination when the variable is regressed against all other explanatory variables. For generalized linear models with categorical variables, we use the Generalized Variance Inflation Factor (GVIF) and its adjusted version $\text{GVIF}^{1/(2\times df)}$, where df is the degrees of freedom of the variable. The criterion for multicollinearity concern is when the adjusted GVIF exceeds 2, which would indicate that more than 80% of the variability can be explained by its dependence on the other explanatory variables.

Variable	**df**	**GVIF**	$\textbf{GVIF}^{1/(2\times df)}$
Age	1	1.307586	1.143498
Sex	1	1.287887	1.134860
Smoking	2	1.178201	1.085449
BMI	1	1.069660	1.034259
Vitamins	2	1.149879	1.072368
Calories	1	13.210244	3.635276
Fat	1	8.175794	2.859334
Fiber	1	2.504249	1.581848
Alcohol	1	2.564752	1.601484
Cholesterol	1	2.195956	1.481875
Beta-carotene	1	1.338719	1.154430

TABLE 12. Explanatory variables with pertaining GVIF and adjusted GVIF values in a model using all 11 explanatory variables.

We notice that the adjusted GVIF of the Calories variable exceeds 2 (specifically 3.635), indicating that more than 80% of its variability can be explained by its dependence on the other explanatory variables (equivalent to $R^2 > 0.80$). Thus, we refit the model given above to not contain the Calories variable and call the new model "Model 3". The new adjusted GVIF values have then been stabilized below the threshold of 2 (i.e., the adjusted GVIF values for Model 3), which may be seen in Table 13.

Variable	**df**	**GVIF**	$\textbf{GVIF}^{1/(2\times df)}$
Age	1	1.219329	1.104232
Sex	1	1.287797	1.134814
Smoking	2	1.177041	1.085004
BMI	1	1.067334	1.033116
Vitamins	2	1.115890	1.056356
Fat	1	2.244437	1.498812
Fiber	1	1.465013	1.210376
Alcohol	1	1.124159	1.059798
Cholesterol	1	2.129296	1.459210
Beta-carotene	1	1.338004	1.156721

TABLE 13. Explanatory variables with pertaining GVIF and adjusted GVIF values in a model using all the 11 explanatory variables except for Calories (i.e., Model 3).

4.4. **d).** We continue with Model 3. In Table 14 the variables are mentioned in connection with their pertaining β-parameters, the β-estimates ($\hat{\beta}$), the e^{β}-estimates ($e^{\hat{\beta}}$) and 95% confidence intervals for the e^{β}-estimates. The categorical variables Smoking, Vitamins and Sex have the reference categories "Current Smoker", "No" and "Female", respectively.

Variable	β	$\hat{\beta}$	$e^{\hat{\beta}}$	**Confidence interval (for $e^{\hat{\beta}}$)**
BMI	β_1	-0.0320	0.9685	(0.9561, 0.9810)
Age	β_2	0.0060	1.0060	(1.0003 1.0117)
Fat	β_3	-0.0012	0.9988	(0.9955, 1.0021)
Cholesterol	β_4	-0.0007	0.9993	(0.9984, 1.0001)
Fiber	β_5	0.0227	1.0230	(1.0058, 1.0405)
Alcohol	β_6	0.0018	1.0018	(0.9954, 1.0083)
Beta-carotene	β_7	0.0559	1.0574	(0.9972, 1.1213)
Smoking (Never)	β_8	0.2729	1.3137	(1.0305, 1.6747)
Smoking (Former)	β_9	0.2010	1.2226	(0.9570, 1.5619)
Sex (Male)	β_{10}	-0.2011	0.8179	(0.6378, 1.0488)
Vitamins (Fairly often)	β_{11}	0.2658	1.3045	(1.0900, 1.5612)
Vitamins (Not often)	β_{12}	0.2671	1.3061	(1.0720, 1.5914)

TABLE 14. Table containing variables with pertaining β-parameters, β-parameter estimates, e^{β}-estimates and 95% confidence intervals for the e^{β}-estimates, for Model 3.

We now perform a series of tests in relation to Model 3. We check whether there is a significant relationship between log beta-carotene levels and BMI (given the other variables in Model 3), if Model 3 is significantly better than Model 1 (which used only BMI) and whether Model 3 is significantly better than Model 2 (which only used Smoking status).

For the first test, since we only want to check the relevance of *one* variable (given all the other variables), this is done with a t-test. For the second test, we want to check whether a subset of the parameters are non-zero (all but the BMI parameter), and thus use a partial F-test. In the third test we want to check if Model 3 is significantly better than Model 2, which only used the Smoking status as explanatory variables. The tests, results and conclusions are shown in Table 15

Test	H_0	**Test statistic**	**Distribution**	**p-value**	**Conclusion**
t	$\beta_1 = 0$	-4.92	student t, 302d.f.	$1.435 \cdot 10^{-6}$	Reject H_0
Partial F	$\beta_i = 0, \forall i \in \{2, 3, ...12\}$	6.2597	F, 302 & 11 d.f.	$2.633 \cdot 10^{-9}$	Reject H_0
Partial F	$\beta_i = 0, \forall i \in \{1, 2, 3, ...12\} \backslash \{8, 9\}$	8.6928	F, 302 & 10 d.f.	$1.632 \cdot 10^{-12}$	Reject H_0

TABLE 15. Table containing test names, null hypotheses, test statistics, the tests' distributions under the null hypotheses, p-values and conclusions for Model 3.

For the first test, we conclude with a p-value of $1.435 \cdot 10^{-6}$ that $\beta_1 \neq 0$ such that there is a significant relationship between log plasma beta-carotene levels and BMI (with $\alpha = 0.01$). This means that a person's BMI most likely affects the levels of beta-carotene in their blood. Again, due to a significantly low p-value ($<< 0.01 = \alpha$), we reject the null hypothesis in test 2; Model 3 is significantly better than Model 1. For the third test, we also decide to reject H_0 with a very low p-value. Hence, it is apparent that Model 3 is a better model than both Model 1 and Model 2.

4.5. **e).** We now inspect the studentized residuals for Model 3. Using figures, we will look for any outliers (abnormally large values of studentized residuals), non-constant variance, and non-normality.

We begin by looking for any outliers. We do this by plotting studentized residuals against the estimate, as seen in Figure 9. We notice several unusually large residuals (those bigger than 2) and two studentized residuals which are suspiciously large. Thus, there are two outliers.

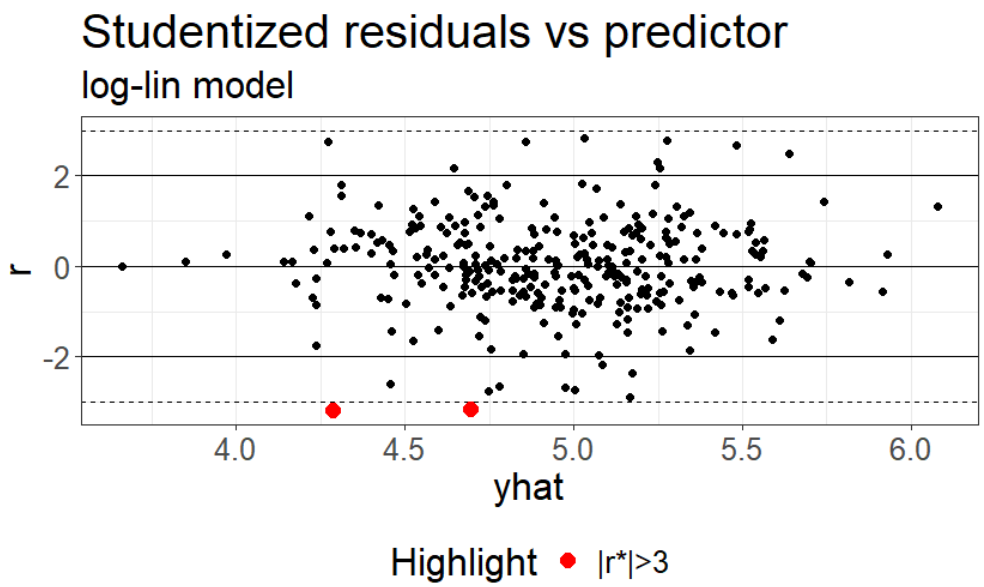


FIGURE 9. Plot of studentized residuals against the linear predictor, highlighting large residuals.

We now want to look for non-constant variance of the studentized residuals. Plotting squared absolute studentized residuals against the linear predictor, as in Figure 10, we find no systematic non-linear pattern. With this in mind, and considering there are only two outliers, we conclude that the residual variance is constant.

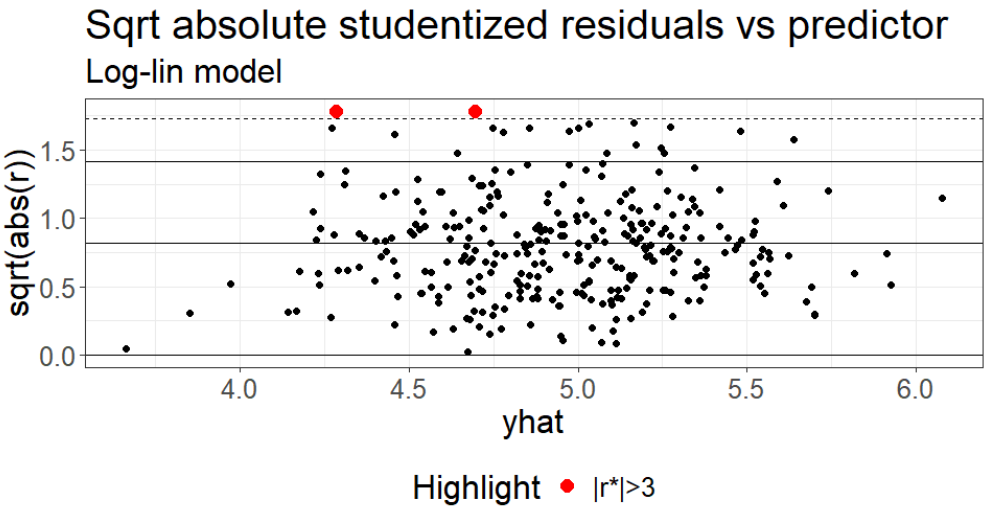


FIGURE 10. Plot of squared absolute studentized residuals against the linear predictor, highlighting large residuals.

4.6. **f).** We now plot the leverage for Model 3 and plot them against the linear predictor, as seen in Figure 11.

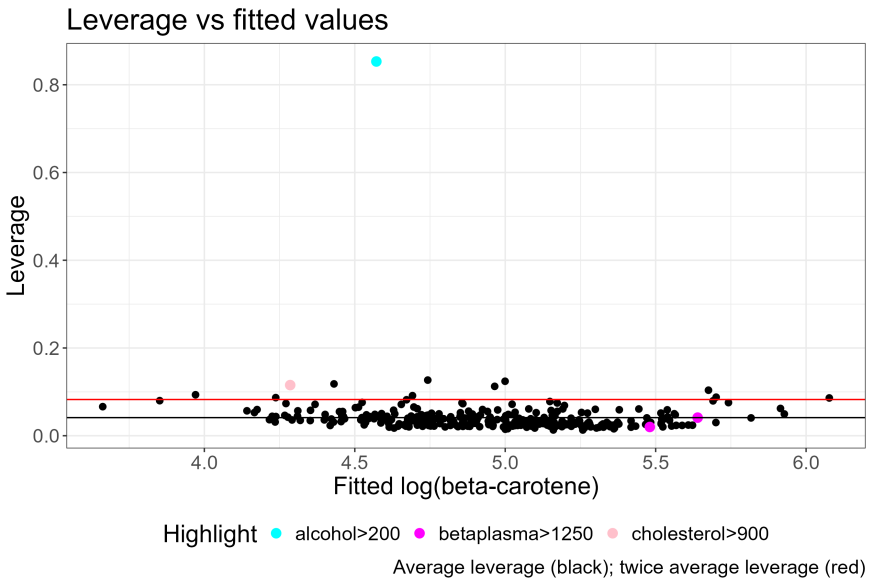


FIGURE 11. Plot of the leverage of Model 3 (including highlights) against linear predictor for the model.

With the help of R, we discover that the observation with largest leverage pertains to the previously mentioned (see section 4.2 (3b)) participant in the study who consumed large amounts of alcohol. This participant naturally also consumed large amount of calories. Due to high alcohol consumption and caloric intake relative to the rest of the participants of the study, the leverage of this observation is very large. In the following two Figures of log beta-carotene levels against alcohol and calories, the extremity of this particular observation becomes very apparent. The figures are given by Figure 12 and Figure 13 respectively, where the outlier has been highlighted.

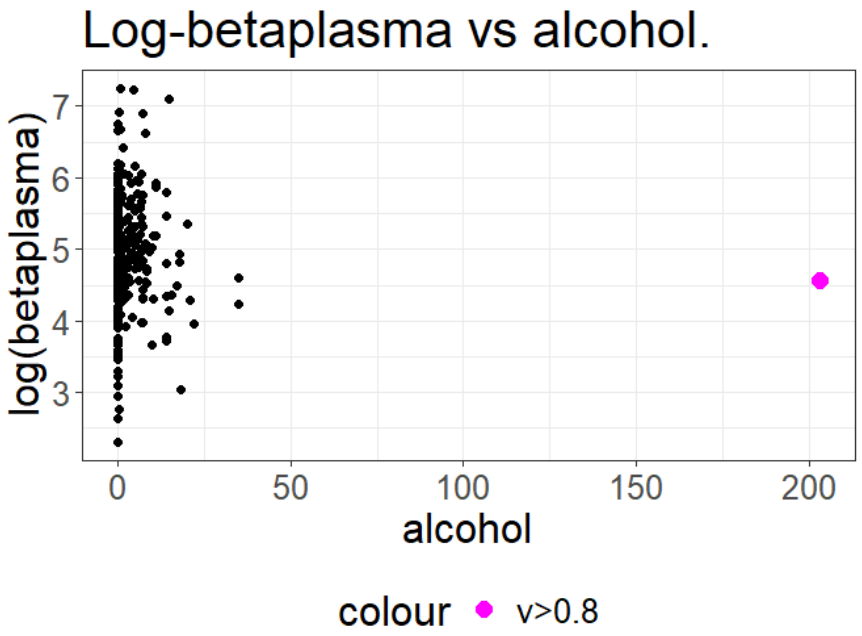


FIGURE 12. Plot of log beta-carotene levels against alcohol, with outlier highlighted.

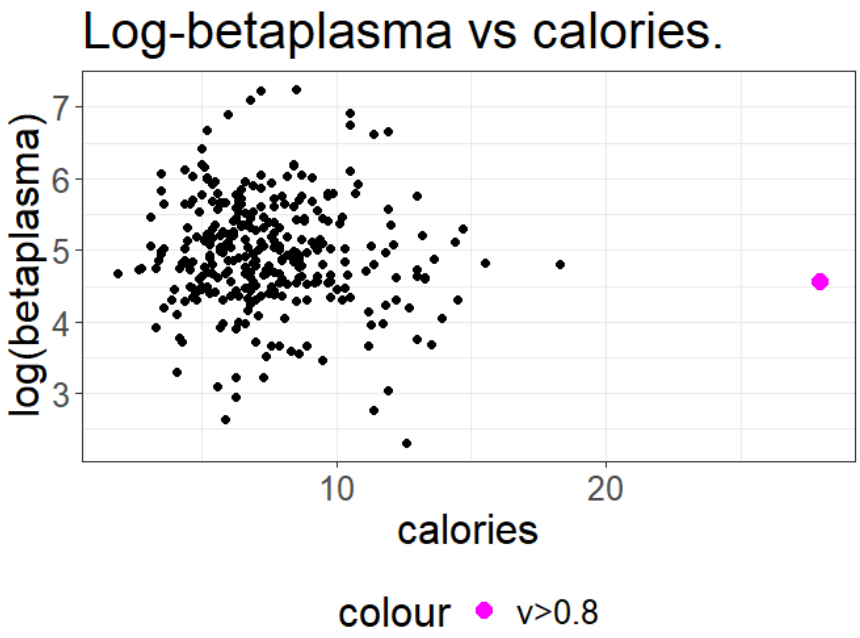


FIGURE 13. Plot of log beta-carotene levels against calories, with outlier highlighted.

The leverage measures the impact of an observation on its own estimated value. Thus, high leverage from outliers *might* skew the model. Thus, the aforementioned abnormal observation may have a high impact on Model 3.

4.7. **g).** We now plot Cook's distance (a measure of influence of observations on a model) for Model 3 against the linear predictor. The plot is given by Figure 14.

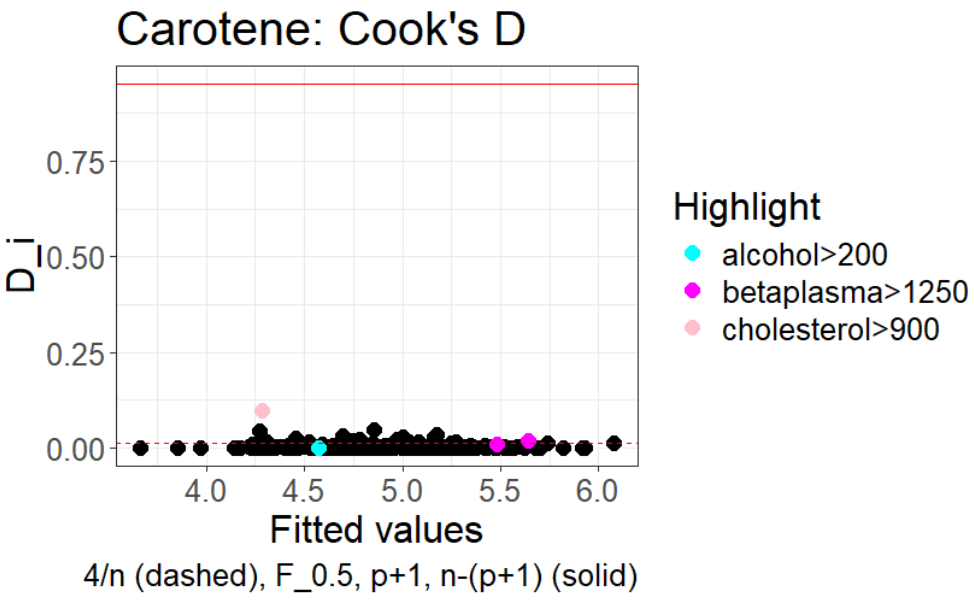


FIGURE 14. Plot of Cook's distance against linear predictor for Model 3, including highlights.

The observation with the highest Cook's distance is *not* the observation with the highest leverage. Rather, it is an observation of a participant with cholesterol consumption higher than 900mg. This is illustrated in Figure 15, where the Cook's distance is plotted against the leverage for the observations.

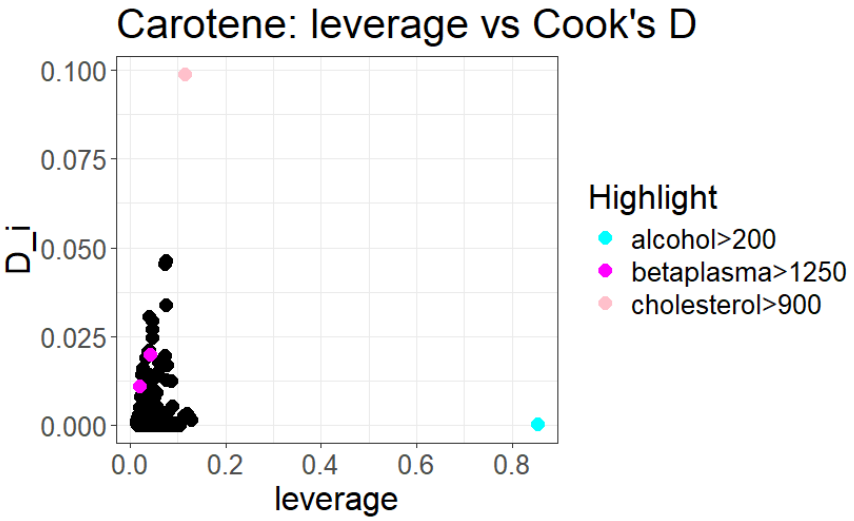


FIGURE 15. Plot of Cook's distance against leverage, including highlights.

After having calculated the DFBETAS for each explanatory variable, we plot the DFBETAS against the fitted values for the variables of interest — the intercept, Fat, Alcohol and Cholesterol. The plot is seen in Figure 16.

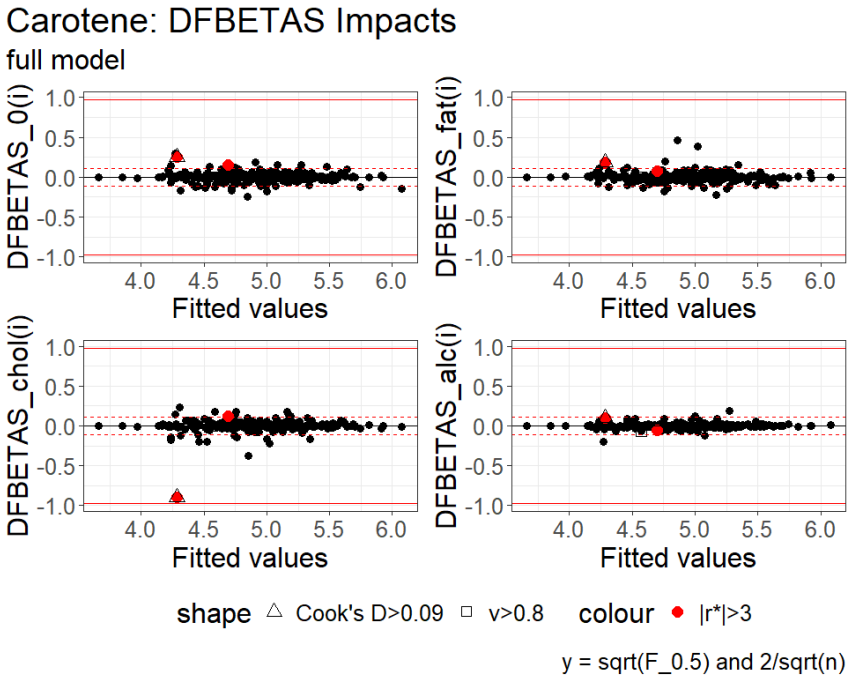


FIGURE 16. Plot of DFBETAS against fitted values for the intercept, Fat, Alcohol and Cholesterol, including highlights.

In relation to the previous discussion about the Cook's distance, we see that the observation with a large Cook's distance in Figure 15 has a large impact on the corresponding β-parameter.

We also note that the observation with the largest leverage (the participant with high alcohol/calorie consumption) did in fact *not* have as high an impact on the model as one could have assumed. This can be seen both in Figure 15 and in Figure 16. An explanation for this could be that the Cholesterol variable has a larger impact than the Alcohol variable, in general.

5. Removing the influential observation

5.1. **a).** We now remove the high impact observation that was found in section 4.7 (3g)) from the data set and estimate a new version of Model 3. In Figure 17, a plot comparing the Cook's distance against the linear predictor, for a Model 3 fitted with and without the highly influential observation is given.

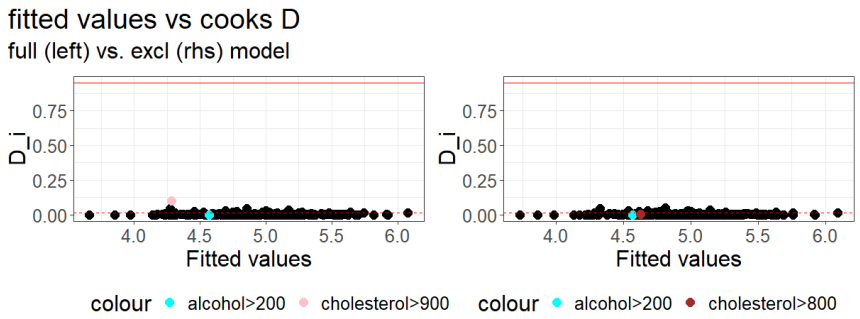


Figure 17. Plot of Cook's distance against linear predictor for Model 3 with and without a highly influential observation, including highlights.

Note that the observation of a participant with a cholesterol consumption of higher than 800mg per day (see plot to the right in Figure 17) does not have a large Cook's distance, despite not consuming much less cholesterol than the participant consuming more than 900mg per day.

The GVIF values for Model 3 on the reduced data set are given in Table 16.

Variable	**GVIF**
Age	1.217516
Sex	1.296542
Smoking	1.177730
BMI	1.064998
Vitamins	1.116915
Fat	2.202732
Fiber	1.464912
Alcohol	1.124967
Cholesterol	2.116094
Beta-carotene	1.345505

Table 16. Explanatory variables with pertaining GVIF-values in a model using all the 11 explanatory variables except for Calories (i.e., Model 3).

Most of the GVIF values are very similar to those of Model 3 with the complete data set, given in Figure 13. The largest differences in the GVIF between them were in the Fat and Cholesterol variables, where the GVIF decreased with -0.042 and -0.0132, respectively.

5.2. **b).** We are to construct two models, one on a full dataset and one on the reduced dataset, through the process of stepwise selection. For $\epsilon_i \in N(0, \sigma^2)$ and $i = 1, ..., n$, the resulting models, referred to as Model 4, are:

$$\log(\text{betaplasma})_i = \beta_0 + \beta_1 \text{bmi}_i + \beta_2 \text{fiber}_i + \beta_3 \text{cholesterol}_i + \beta_4 \text{vituse}_{1i} + \beta_5 \text{vituse}_{2i} + \epsilon_i \tag{5.1}$$

and

$$\log(\text{betaplasma})_i = \beta_0 + \beta_1 \text{bmi}_i + \beta_2 \text{fiber}_i + \beta_3 \text{calories}_i + \beta_4 \text{vituse}_{1i} + \beta_5 \text{vituse}_{2i} + \epsilon_i, \tag{5.2}$$

One can observe that there was a significant difference; when constructing model on the full dataset we had to include 'cholesterol', while on the model that had the cholesterol outlier excluded 'calories' was the variable that was preferred by the model selection procedure. The order of one-by-one variable addition to a null model in the process of stepwise variable selection for the full model was: bmi, fiber, cholesterol, vituse; wheareas on the other model: bmi, fiber, calories vituse.

This is interesting, and the difference seems reasonable. Before, we have established that the outlier observation likely has an impact on model fit. And now, when the components of the model have to have higher explanatory power (as we have fewer of them and only the most significant ones), we see the impact of this one observation is significant to an extent that is able to influnce BIC values of related component greatly.

All of the coefficients are of high significance. See coefficients and confidence intervals below (mind the small values due to log scale):

	Beta	**2.5%**	**97.5%**
(Intercept)	5.39460	4.97575	5.81345
BMI	-0.02913	-0.04195	-0.01632
Fiber	0.03222	0.01764	0.04681
Cholesterol	-0.00120	-0.00179	-0.00062
Vitamin Use (Yes, fairly often)	0.32591	0.14787	0.50394
Vitamin Use (Yes, not often)	0.27643	0.08024	0.47263

TABLE 17. Full Model Regression Coefficients

	Beta	**2.5%**	**97.5%**
(Intercept)	5.40728	4.98396	5.83061
BMI	-0.03006	-0.04261	-0.01751
Fiber	0.04072	0.02469	0.05676
Calories	-0.05199	-0.08190	-0.02208
Vitamin Use (Yes, fairly often)	0.34923	0.17352	0.52493
Vitamin Use (Yes, not often)	0.27986	0.08671	0.47301

TABLE 18. On Excl Data Regression Coefficients

6. Fine-tuning the model

6.1. **a).** Now we seek to remove some categories in categorical variable ('vituse') in the Model 4. After some experimentation, the best option was to group the 'vituse' factor variable in a wat so it only has two levels: Yes (combines what used to be 'Yes, not often', 'Yes, fairly often') and 'No'. Different grouping did not produce better results.

We would like to see if there are any differences between this model, Model 5, and the Model 4 on a reduced data set. And we see there are no significant differences in the beta coefficient estimates between models 4 and 5:

	Beta	**2.5%**	**97.5%**
(Intercept)	5.39902	4.98104	5.81701
BMI	-0.02935	-0.04212	-0.01658
Fiber	0.03241	0.01787	0.04696
Cholesterol	-0.00121	-0.00179	-0.00062
vit_ynYes	0.30568	0.14628	0.46508

TABLE 19. Full Model Regression Coefficients

	Beta	**2.5%**	**97.5%**
(Intercept)	5.41503	4.99257	5.83748
BMI	-0.03038	-0.04289	-0.01787
Fiber	0.04107	0.02508	0.05707
Calories	-0.05245	-0.08231	-0.02259
vit_ynYes	0.32069	0.16355	0.47784

TABLE 20. Exclusion Model Regression Coefficients

It is worth adding that it is not necessary to change the 'vituse' factor variable in this way (ANNOVA does not see big difference between models, the adjusted R squared is not much improved). However, given lesser complexity of the model, it makes sense to keep this new version.

6.2. **b).** Now we compare the five models with each other (1, 2, 3, 4, 5)–we look at the number of parameters, the residual standard deviation σ, R^2 and adjusted R^2, degrees of freedom, as well as the AIC and BIC metrics. It is safe to say that the last model, Model 5 seems to be the most rational choice, given its performance and number of parameters. It has the lowest BIC value (see table bellow), slightly outperforming 4. The largest model, Model 3 may have most explanatory power and lowest residual standard deviation, yet it is perhaps over-parametrized as a model with only one coefficient, 1, even has a higher BIC. Simplistic models, 1 and 2, are expected to be worse; smoking seems to have very poor explanatory power.

	Name	Number of β	σ	R2	R2 adjusted	df	AIC	BIC
1	Model 5	5	0.6729648	0.2002493	0.1898966	6	649.3261	671.8224
2	Model 4	6	0.6734953	0.2015802	0.1886189	7	650.8031	677.0489
3	Model 1	2	0.7189481	0.0783605	0.0754065	3	687.8683	699.1165
4	Model 3	13	0.6618880	0.2463894	0.2163452	14	646.6669	699.1584
5	Model 2	3	0.7348585	0.0402033	0.0340309	4	702.6064	717.6040

TABLE 21. Model Summary Statistics

7. AI DISCLOSURE

Uses of ChatGPT: code command lookup and debugging (R, Latex) , plot formatting (R, Latex).

8. Project contributions

Kseniia Sharnina: Analysis, Discussions, Programming, Visualisation. Task section 3 code (mostly the latter parts are used), sections 4-5 code (except parts of Konstantinos's code 4(a)); writing 4(b)-5. Revision of code and revision of text for sections 1 and 2 of tasks.

Leonard Nygren Löhndorf: Revised groupmates codes. Writing and revising first draft of Introduction and section 1a-4a (numbered as 1-5.1 in the document) based on groupmates' codes. This includes creating figures, tables, analysis of code, formatting, writing, etc..

Konstantinos Vitos: Programming, Visualisation, Analysis. Sections 1-3 (fixxed some errors spotted by Kseniia).